Electronic plants in the making?

Augmenting plants with electronic functionality might make it possible to combine electric signals with the plants' own electrical signals, based on their chemical processes. These technologies can not only be used as tools by plant biologists but can also find applications in agriculture and forestry, with today's climate change and increasing population. According to researchers, both analogue and digital electronic circuits can be created inside living plants, be it flowers, bushes or trees, with the use of semi-conductive polymers.

Integrating electronics in plants can also play an important role towards bio-hybrid systems, plant optimisation and monitoring. Research efforts are going on in the field of template electronics to understand how plant biochemistry is affected by electronics and to make plants a part of our technology. Laboratory stage studies on plant growth in a controlled environment and development, characterisation, and integration of electronic devices and materials in plants are also underway.

The science behind plants

The various parts of a plant such as root, stem, leaves, and vascular circuitry are responsible for conveying chemical signals as well as regulating the growth and functions of a plant. From a certain viewpoint, these features are said to be equivalent to the contacts, interconnections, and wires of discrete and integrated electronic circuits and devices.

Several attempts have been made so far to enhance plant functions with the use of electro-active materials but without any efforts to directly merge electronics with a plant's circuitry. The tree-power activity is totally different to the popular potato or lemon experiment, in which two different metals react with the food to create an electric potential difference that causes a current to flow.

Can trees power circuits?

A custom circuit can be designed to store enough voltage from trees to run a low-power sensor. Scientists have found that by putting one electrode in a plant and the other in the surrounding soil, plants can generate a voltage of up to 200mV. The research team wanted to advance their research in the field of generating power from trees by building circuits to run off plant electric energy and successfully ran a custom circuit solely off tree power.

It is not established so far as to where the voltages in plants come from, but there seems to be some signaling in trees similar to what happens in the human body, while at slower speeds. Traditional electronics can send and process electronic signals whereas plants transport and handle ions and growth hormones. In case of organic electronics based on semi-conductive polymers, both ions and electrons can serve as signal carriers, which make it possible to combine electric signals with the plant's own electrical signals, as if